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## 1 Day 1. Classic binomial and multinomial identities

### Planned entrance quiz :)

1. How many ways are to order 6 persons in a queue?
2. The box contains 50 balls numbered from 1 to 50. We draw 6 balls without replacement and write down the resulting sequence of numbers. How many different sequences are possible?
3. I have 5 yellow books and 7 red books. I want to place them on my bookshelf and I would like to place yellow books to the left of red books. How many book orders are possible?

**Definition 1.1.** Factorial of  $n$ , denoted by  $n!$ , is the number of ways to put  $n$  distinct objects in a queue.

**Definition 1.2.** Binomial coefficient  $\binom{n}{k}$  is the number of ways to select  $k$  distinct objects from  $n$  distinct objects.



**Definition 1.3.** For  $n = a + b + c$  the multinomial coefficient  $\binom{n}{a,b,c}$  is the number of ways to split  $n$  distinct objects into different boxes, where the first box contains  $a$  objects, the second box contains  $b$  objects, the third box contains  $c$  objects.

One may create multinomial coefficient for any number of boxes :)

**Exercise 1.1.** Is  $\binom{n}{k}$  equal to  $\binom{n}{n-k}$ ?

Is  $\binom{n}{k}$  equal to  $\binom{n}{k,n-k}$ ?

**Exercise 1.2.** Write the product

$$\binom{100}{20} \cdot \binom{80}{50} \cdot \binom{30}{10}$$

as one multinomial coefficient.

**Exercise 1.3.** Write

$$\binom{n}{a,b,c,d}$$

using binomial coefficients.

Write

$$\binom{n}{a,b,c,d}$$

using factorials of  $n, a, b, c$  and  $d$ .

**Idea 1.** Solution strategies for combinatorial identities:

1. Double counting or story proof.
2. Visual identity (rare gem!).
3. Evaluation of a polynomial at a special point. Differentiation, integration.
4. Combine old identities into new.
5. Guess the answer and verify by induction.
6. Dark art!

## Pascal's triangle and rule

**Exercise 1.4.** Recall the Pascal arithmetic triangle. Draw two versions: triangular and rectangular.

**Exercise 1.5.** Pascal's rule. Simplify

$$\begin{aligned} & \binom{n}{k} + \binom{n}{k+1}. \\ & \binom{n}{a-1,b,c} + \binom{n}{a,b-1,c} + \binom{n}{a,b,c-1}. \\ & \binom{n}{a-1,b,c,d} + \binom{n}{a,b-1,c,d} + \binom{n}{a,b,c-1,d} + \binom{n}{a,b,c,d-1}. \end{aligned}$$



**Exercise 1.6.** Simplify

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

Recall the Pascal's rule and simplify

$$\binom{2n}{0} + \binom{2n}{2} + \binom{2n}{4} + \cdots + \binom{2n}{2n}.$$

Simplify

$$\binom{2n-1}{0} + \binom{2n-1}{2} + \binom{2n-1}{4} + \cdots + \binom{2n-1}{2n-2}.$$

**Exercise 1.7.** Simplify

$$\begin{aligned} & \sum_{a+b+c=2026} \binom{2026}{a, b, c}. \\ & \sum_{a+b+c+d=2027} \binom{2027}{a, b, c, d}. \\ & \sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}. \\ & \sum_{a+b \leq n} \binom{n}{a} \cdot \binom{n-a}{b}. \end{aligned}$$

Here  $a$ ,  $b$  and  $c$  are assumed non-negative integers.

### Hockey-stick identity

**Exercise 1.8.** Hockey-stick identity , Christmas stocking identity, boomerang identity.  
Simplify

$$\begin{aligned} & \sum_{n=a}^b \binom{n}{a}. \\ & 1 + 2 + 3 + \cdots + n. \\ & 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n \cdot (n+1). \\ & 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \cdots + n \cdot (n+1) \cdot (n+2). \end{aligned}$$

**Exercise 1.9.** Obtain closed form expressions for:

$$\begin{aligned} & 1^2 + 2^2 + \cdots + n^2, \\ & 1^3 + 2^3 + \cdots + n^3, \\ & 1^4 + 2^4 + \cdots + n^4. \end{aligned}$$

**Exercise 1.10.** Hockey-sticks plus idea from linear algebra!

a) Decompose  $n^2 + 2n + 5$  using  $\binom{n}{0}$ ,  $\binom{n}{1}$ ,  $\binom{n}{2}$  as the basis.

b) Find the sum

$$\sum_{k=2}^{2026} k^2 + 2k + 5.$$

c) Using similar trick find the sums

$$\sum_{k=1}^n 5k^2 + 4k, \quad \sum_{k=1}^n k^3 + 2k^2.$$

**Exercise 1.11.** Let's start from some number  $\binom{n}{k}$  inside the Pascal's triangle.  
In one step we can go either in north-east or north-west direction.  
Find the sum of all numbers we can reach.

### Binomial or Taylor or MacLaurin expansion

**Exercise 1.12.** Write the Taylor expansion (binomial expansion) at  $x = 0$  for

$$(1+x)^n, \quad (x+y)^n.$$

**Exercise 1.13.** By plugging some special points to the  $(x+y)^n$  or somewhere else find the sums

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$
$$\sum_{k=0}^n k \binom{n}{k}, \quad \sum_{k=0}^n (k-1) \binom{n}{k}, \quad \sum_{k=0}^n (-1)^k \binom{n}{k}, \quad \sum_{k=0}^n (-1)^{k-1} k \binom{n}{k}$$

Hint: differentiation may help you.

### How that can be?

**Exercise 1.14.** Be brave, really brave!

a) Take a deep breath!

b) Calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

**Exercise 1.15.** a) Simplify

$$\binom{-1/2}{3} + \binom{-1/2}{4}.$$

b) Draw the Pascal's triangle (the whole half-plane!) starting from  $\binom{-1/2}{0}$ .

c) Draw the Pascal's triangle (the whole half-plane!) starting from  $\binom{1/3}{0}$ .

**Exercise 1.16.** Write the Taylor expansion at  $x = 0$  for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

**Exercise 1.17.** Simplify

$$\binom{10}{0} \binom{-0.5}{7} + \binom{10}{1} \binom{-0.5}{6} + \binom{10}{2} \binom{-0.5}{5} + \cdots + \binom{10}{7} \binom{-0.5}{0}.$$

Hint: for a honest proof consider the binomial expansion of  $(1+x)^1$ ,  $(1+x)^{-0.5}$ ,  $(1+x)^{10-0.5}$ .

**Exercise 1.18.** Homage to Newton.

- a) Write the Taylor expansion at  $x = 0$  for  $(1 - x^2)^{1/2}$ .  
b) Visually (or analytically if you wish) find the integral

$$\int_{-1}^1 (1 - x^2)^{1/2}.$$

- c) Calculate the sums

$$\binom{1/2}{0} - \frac{1}{3} \binom{1/2}{1} + \frac{1}{5} \binom{1/2}{2} + \cdots$$
$$\sum_{k=2}^{\infty} \frac{1}{2k+1} \frac{(2k-3)!!}{(2k)!!}$$

**Home assignment-1:**

1. Simplify

$$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$

2. 1.9(2) in the old enumeration:

Find explicit formula for the sum

$$1^3 + 2^3 + \cdots + n^3.$$

Hint: represent the general term  $k^3$  as the sum of binomial coefficients and then use the hockey stick or Christmas stocking or boomerang!

3. [\*] Using hockey stick decomposition find the sums

$$\sum_{k=1}^n 5k^2 + 4k.$$

4. [\*] Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

5. Write the Taylor expansion at  $x = 0$  for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

6. Watch a Veritasium video «Newton changed the game», [Ver21].

<https://youtu.be/gMlf1ELvRzc?si=VASoVKQj3Pk-wstA>,



You are more than welcome to solve other exercises that we skipped. Starred exercises will be on the quiz next day.

## 2 Day 2. Stars, bars and Gauss summation trick

### Day 2 welcome quiz!

1. Decompose the function  $h(k) = 2k^2 + k$  using the binomial coefficient basis  $\binom{k}{0}$ ,  $\binom{k}{1}$  and  $\binom{k}{2}$ .
2. Let's play hockey! Find the sum

$$\sum_{k=2}^{1000} 3\binom{k}{0} - 2\binom{k}{1} + 3\binom{k}{2}.$$

Hint: do not forget to bring money back to the bank if you borrow!

3. Find the value

$$\binom{1/3}{2} + \binom{1/3}{3} + \binom{4/3}{2}.$$

### Gauss summation trick

**Exercise 2.1.** Simplify

$$\begin{aligned} & 1 + 2 + 3 + \cdots + n; \\ & 1^2 + 2^2 + 3^2 + \cdots + n^2; \\ & \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}; \\ & \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n}; \\ & \binom{n}{2} + 2\binom{n}{3} + 3\binom{n}{4} + \cdots + (n-1)\binom{n}{n}; \\ & \binom{n}{0} + 3\binom{n}{1} + 5\binom{n}{2} + \cdots + (2n+1)\binom{n}{n}; \\ & \binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} + \cdots + (-1)^n(n+1)\binom{n}{n}; \\ & 3\binom{n}{1} + 7\binom{n}{2} + 11\binom{n}{3} + \cdots + (4n-1)\binom{n}{n}; \\ & \binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - 4\binom{n}{4} + \cdots + (-1)^n n\binom{n}{n}; \end{aligned}$$

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## Stars and bars

**Exercise 2.2.** There are 20 chairs in a row and 7 students. Each student needs one chair to sit down. We do care only about which chairs are occupied.

- a) In how many ways chairs can be occupied?
- b) In how many ways chairs can be occupied if there should be at least one empty chair between occupied chairs?
- c) In how many ways chairs can be occupied if there should be at least two empty chairs between occupied chairs?

**Exercise 2.3.** I have four bags of different colors and 20 identical marble stones.

- a) In how many different ways can I put marbles into bags?
- b) In how many different ways can I put marbles into bags if there should be no empty bags?

**Theorem 2.1** (Stars and bars theorem). The number of ways to distribute  $n$  identical objects into  $k$  distinguishable bags is equal to

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}.$$

Hint:  $n$  is the number of stars and  $(k-1)$  is the number of bars.

**Exercise 2.4.** There are 5 types of post-cards sold at the post-office.

- a) In how many different ways can I buy 12 post-cards?
- b) In how many different ways can I buy 12 post-cards if I would like to buy at least one post-card of every type?

**Exercise 2.5.** Consider the equation  $x_1 + x_2 + x_3 + x_4 = 2026$ .

- a) How many solutions does it have if  $x_1, x_2, x_3$  and  $x_4$  are non-negative integers?
- b) How many solutions does it have if  $x_1, x_2, x_3$  and  $x_4$  are positive integers?
- c) How many solutions does it have if all  $x_i$  are integers and  $x_i \geq 2$  for all  $i$ ?
- d) How many solutions does it have if all  $x_i$  are integers and  $x_i \geq i$  for all  $i$ ?

**Exercise 2.6.** There are 10 standard indistinguishable dices.

- a) How many different combinations of scores are possible?
- b) How many different combinations of scores are possible if at least one dice shows 6?
- c) How many different combinations of scores are possible if no dice shows 5 nor 6?

**Exercise 2.7.** Stars, bars, circles and probability!

Consider a well shuffled deck of 52 cards. I open the cards one by one.

- a) How many cards on average I open to see the first Queen?
- b) How many cards on average I open to see the third Ace?
- c) What is expected number of cards between the third and the fourth King?
- d) What is the probability that the third Ace will be the 20-th card opened?
- e) Write combinatorial identities hidden in a), b) and c).

## Visualize it

**Exercise 2.8.** a) Express as a binomial coefficient and visualize the sum:

$$1 + 2 + 3 + 4 + \cdots + n.$$

- b) Visualize the Fuji-sum and simplify it!

$$1 + 2 + 3 + \cdots + (n - 1) + n + (n - 1) + \cdots + 3 + 2 + 1.$$

- c) Recall the multiplication table for numbers from 1 to 10. Why the sum of all numbers is just

$$1^3 + 2^3 + 3^3 + \cdots + 10^3?$$

Simplify the sum!

## Home assignment: day 2 → day 5

Deadline: 2026-07-03, 16:59.

1. There are 10 standard indistinguishable dices.
  - a) How many different combinations of scores are possible?
  - b) How many different combinations of scores are possible if at least one dice shows 6?
  - c) How many different combinations of scores are possible if no dice shows 5 nor 6?
2. There are 5 types of post-cards sold at the post-office.
  - a) In how many different ways can I buy 12 post-cards?
  - b) In how many different ways can I buy 12 post-cards if I would like to buy at least one post-card of every type?
3. Using Gauss summation trick or otherwise find the sum:

$$\binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} \cdots + (-1)^n(n+1)\binom{n}{n};$$

4. Recall the mother-identity

$$(1+x)^n = 1 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

(a) Find the first derivative of the mother-identity.

(b) Find the value of the sum

$$\binom{n}{1} + 2 \cdot 7 \cdot \binom{n}{2} + 3 \cdot 7^2 \cdot \binom{n}{2} + \cdots + n \cdot 7^{n-1} \binom{n}{n}.$$

(c) Find the second derivative of the mother-identity.

(d) Find the value of the sum

$$2 \cdot 1 \binom{n}{2} + 3 \cdot 2 \cdot 9 \cdot \binom{n}{3} + 4 \cdot 3 \cdot 9^2 \cdot \binom{n}{4} + 5 \cdot 4 \cdot 9^3 \cdot \binom{n}{5} + \cdots + n(n-1)9^{n-2} \binom{n}{n}.$$

### Demo for the quiz on day 3

1. Using Gauss summation trick or otherwise find the sum

$$3 \binom{n}{1} + 5 \binom{n}{2} + 7 \binom{n}{3} + \cdots + (2n+1) \binom{n}{n}.$$

2. Consider the equation  $a + b + c = 100$  with integers  $a \geq 3, b \geq 2, c \geq 5$ .

How many solutions are there?

## 3 Day 3. Recurrence relations

**Exercise 3.1.** Recurrence equations starter pack :)

a) Solve the recurrence equation:

$$a_n = a_{n-1} + 3, \quad a_1 = 20.$$

b) Solve the recurrence equation:

$$b_n = 3 \cdot b_{n-1}, \quad b_1 = 20.$$

c) Solve the recurrence equation:

$$c_n = (n+3) \cdot c_{n-1}, \quad c_1 = 20.$$

d) Solve the recurrence equation:

$$d_n = n + 3 + d_{n-1}, \quad d_1 = 20.$$

e) Solve the recurrence equation:

$$e_n = n^2 - 3n + 2 + e_{n-1}, \quad e_1 = 20.$$

f) Solve the recurrence equation:

$$f_n = (n^2 - 3n + 2)f_{n-1}, \quad f_1 = 20.$$

g) Solve the recurrence equation:

$$g_n = \frac{n+2}{n+1} g_{n-1}, \quad g_1 = 20.$$

h) Solve the recurrence equation:

$$h_n = \frac{1}{n^2 + n} + h_{n-1}, \quad h_1 = 20.$$

**Exercise 3.2.** Gauss trick with subtraction!

a) Find the sum

$$S = 2 + 2 \cdot (1/3) + 2 \cdot (1/3)^2 + 2 \cdot (1/3)^3 + \dots$$

b) Solve the recurrence equation

$$S_n = S_{n-1} + 3 \cdot (1/5)^n, \quad S_0 = 0.$$

c) Solve the recurrence equation

$$S_n = S_{n-1} + n \cdot (1/5)^n, \quad S_0 = 0.$$

**Exercise 3.3.** You draw  $n$  lines on a plane, they split the plane into some number of regions. Lines may or may not coincide.

a) What is the minimal number of regions  $a_n$ ?

b) What is the maximal number of regions  $b_n$ ?

Hint: start by finding  $b_1, b_2$ , then find the recurrence equation for  $b_n$ .

**Idea 2.** Consider the equation

$$x_t + \alpha_1 x_{t-1} + \alpha_2 x_{t-2} = \text{poly}(t).$$

The solution can be always represented as a sum or product of some geometric series and polynomials.

**Exercise 3.4.** Unique root case.

a) Solve the recurrence equation:

$$a_n - 5a_{n-1} + 6a_{n-2} = 0, \quad a_0 = 3, a_1 = 5.$$

b) Solve the recurrence equation:

$$a_n - 7a_{n-1} + 12a_{n-2} = 6, \quad a_0 = 1, a_1 = 12.$$

c) Solve the recurrence equation:

$$a_n - 8a_{n-1} + 15a_{n-2} = 8n - 22, \quad a_0 = 5, a_1 = 22.$$

**Exercise 3.5.** We start with one pair of young rabbits  $F_1 = 1$ . It takes one period of time for a pair of rabbits to mature. Every mature pair of rabbits in one period of time will give birth to a new young pair of rabbits.

a) Find  $F_2, F_3, F_4, F_5$ .

b) Deduce the recurrence relation for  $F_n$ .

c) Find the general formula for  $F_n$ .

d) Which term in the general formula can be ignored for the big  $n$ ?

e) Guess and prove formulas for

$$\sum_{k=0}^n F_k, \quad \sum_{k=0}^n F_{2k}, \quad \sum_{k=1}^n F_{2k-1}, \quad F_{n+1}F_{n-1} - F_n^2.$$

f) Which of the following formulas are valid?

$$F_{m+n} = F_m F_{n+1} + F_{m-1} F_n$$

$$F_n^2 + F_{n+1}^2 = F_{2n+1}$$

$$F_{n+1}^2 - F_{n-1}^2 = F_{2n}$$

$$\sum_{k=0}^n F_k^2 = F_n F_{n+1}.$$

**Exercise 3.6.** The mighty lag operator!

Let  $L$  be the lag operator and  $\Delta = 1 - L$ . Consider the sequences  $x_n = 5^n$ ,  $y_n = n \cdot 5^n$  and  $z_n = 3^n$ .

- Find  $(1 - 3L)x_n$ ,  $\Delta x_n$ ,  $(1 - 5L)x_n$ .
- Find  $\Delta(1 - 3L)x_n$ ,  $\Delta(1 - 5L)x_n$ ,  $(1 - 5L)^2 x_n$ .
- Find  $(1 - 3L)y_n$ ,  $\Delta y_n$ ,  $(1 - 5L)y_n$ .
- Find  $\Delta(1 - 3L)y_n$ ,  $\Delta(1 - 5L)y_n$ ,  $(1 - 5L)^2 y_n$ .
- Find  $(1 - 3L)(1 - 5L)x_n$  and  $(1 - 3L)(1 - 5L)z_n$ .
- Rewrite the equation  $a_t - 7a_{t-1} + 12a_{t-2} = 0$  using lag operator.

Warning! Remember that  $(1 - 3L)$  or  $(1 - 5L)$  is not a number! Lag operator is just a convenient notation to shift index!

**Exercise 3.7.** Double root case.

- Solve the recurrence equation:

$$a_n - 8a_{n-1} + 16a_{n-2} = 0, \quad a_0 = 3, a_1 = 0.$$

- Solve the recurrence equation:

$$a_n - 4a_{n-1} + 13a_{n-2} = 20, \quad a_0 = 6, a_1 = 4.$$

- Solve the recurrence equation:

$$q_n - 6q_{n-1} + 9q_{n-2} = 8n - 20, \quad q_0 = 2, a_1 = 12.$$

## Home assignment: day 3 → day 6

Deadline: 2026-07-06, 14:59.

1. Solve the recurrence equation:

$$a_n = (n + 1) + a_{n-1}, \quad a_1 = 100.$$

Hint: recall that  $1 + 2 + 3 + 4 + \dots + n = \binom{n+1}{2} = \frac{n(n+1)}{2}$ .

2. Solve the recurrence equation:

$$a_n + 5a_{n-1} + 6a_{n-2} = 0, \quad a_0 = 3, a_1 = 5.$$

3. Solve the recurrence equation:

$$c_n = 2 \cdot c_{n-1} + 3, \quad c_1 = 20.$$

4. Solve the recurrence equation:

$$a_n - 7a_{n-1} + 12a_{n-2} = 6, \quad a_0 = 1, a_1 = 12.$$

5. Watch a Veritasium video «This equation will change how you see the world», [Ver20].

<https://youtu.be/ovJcsL7vyrk?si=kh8wjM2e6dgLSXzj>.



## Demo for the quiz on day 4

1. Solve the recurrence equation:

$$b_n = \frac{n+5}{n+4}b_{n-1}, \quad b_1 = 100.$$

2. Here  $L$  is the lag operator and  $x_n = 6^n$ . Find the values

$$(1 - 5L)x_n, \quad (1 - 6L)x_n, \quad (1 - 6L)(1 - 5L)x_n.$$

## 4 Day 4. More recurrence equations

**Exercise 4.1.** Double root case.

- a) Solve the recurrence equation:

$$a_n - 8a_{n-1} + 16a_{n-2} = 0, \quad a_0 = 3, a_1 = 0.$$

- b) Solve the recurrence equation:

$$a_n - 4a_{n-1} + 4a_{n-2} = 20, \quad a_0 = 20, a_1 = 21.$$

- c) Solve the recurrence equation:

$$q_n - 6q_{n-1} + 9q_{n-2} = 8n - 20, \quad q_0 = 2, a_1 = 12.$$

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**Exercise 4.2.** Unique complex root case.

a) Solve the recurrence equation:

$$a_n - 4a_{n-1} + 13a_{n-2} = 0, \quad a_0 = 4, a_1 = 2.$$

b) Solve the recurrence equation:

$$a_n - 4a_{n-1} + 13a_{n-2} = 20, \quad a_0 = 6, a_1 = 4.$$

c) Solve the recurrence equation:

$$a_n + 9a_{n-2} = 10n - 8, \quad a_0 = 7, a_1 = 8.$$

**Exercise 4.3.** I enter a casino in Monte Carlo with 50 tokens. I play the roulette. I always bet one token on black. I will leave the casino when my fortune will be 70 tokens or 0 tokens.

In a typical Monte Carlo roulette there are 18 black pockets, 18 red pockets and zero. Let  $p_n$  be the probability that I will leave the casino as winner if I have  $n$  tokens now.

- a) What are the initial conditions  $p_0$  and  $p_{70}$ ?
- b) Build the recurrence equation for  $p_n$ .
- c) Solve the recurrence equation and find  $p_{50}$ .
- d) How the answers would change for Las Vegas roulette with zero and double zero pockets?
- e) How the answers would change for an abstract fair casino with no zero nor double zero?

Let  $e_n$  be the expected number of spins before I leave the casino (as winner or as loser).

- 6. What are the initial conditions  $e_0$  and  $e_{70}$ ?
- 7. Build the recurrence equation for  $e_n$ .
- 8. Solve the recurrence equation and find  $e_{50}$ .
- 9. How the answers would change for Las Vegas roulette with zero and double zero pockets?
- 10. How the answers would change for an abstract fair casino with no zero nor double zero?

**Exercise 4.4.** I have a broken light switch button in my room. The switch button should turn the light on if the light is off and vice versa. On the first press the button works with probability  $1/2$ , on the second press — with probability  $1/3$ , on the third one — with probability  $1/4$  and so on. Initially my room is dark.

What is the probability that my room will be dark after I press the button 100 times?

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## Home assignment: day 4 → day 7

Deadline: 2026-07-07, 14:59.

1. Solve the recurrence equation:

$$a_n + 6a_{n-1} + 9a_{n-2} = 0, \quad a_0 = 5, a_1 = 10.$$

2. Solve the recurrence equation:

$$c_n + 6c_{n-1} + 10c_{n-2} = 0, \quad c_0 = 0, c_1 = 10.$$

3. Solve the recurrence equation:

$$a_n = (n + 5)a_{n-1}, \quad a_1 = 100.$$

You may certainly use factorial in your answer!

4. I enter a casino in Las Vegas with 50 tokens. I play the roulette. I always bet one token on black. I will leave the casino when my fortune will be 70 tokens or 0 tokens.

In a typical Las Vegas roulette there are 38 pockets: 18 black pockets, 18 red pockets, one zero and one double zero. Let  $p_n$  be the probability that I will leave the casino as winner if I have  $n$  tokens now.

- What are the initial conditions  $p_0$  and  $p_{70}$ ?
- Build the recurrence equation for  $p_n$ .
- Solve the recurrence equation for  $p_n$ .
- Evaluate  $p_{50}$  up to 2 digits.

## Demo for the quiz on day 5

1. Consider the recurrence equation  $y_n - 7y_{n-1} + 10y_{n-2} = 0$ .

Find all solutions of the form  $y_n = \lambda^n$ .

2. You know that  $r_n = c \cdot 2^n + d \cdot n \cdot 2^n + 4$  is the solution of some recurrence equation with initial conditions  $r_0 = 6$  and  $r_1 = 10$ .

Find  $c$  and  $d$ .

## 5 Day 5. Systems of linear recurrence equations, examples from probability

**Exercise 5.1.** Now the systems!

- a) Solve the recurrence equation:

$$\begin{cases} a_n = 3a_{n-1} + 2b_{n-1} - 4, \\ b_n = 2a_{n-1} + 3b_{n-1} - 4, \\ a_0 = 1, \quad b_0 = 0. \end{cases}$$

b) Solve the recurrence equation:

$$\begin{cases} a_n = 3a_{n-1} - 2b_{n-1} - 4, \\ b_n = -2a_{n-1} + 3b_{n-1} + 4, \\ a_0 = 1, \quad b_0 = 0. \end{cases}$$

**Exercise 5.2.** There are  $n$  passengers boarding an intercity bus and standing in queue. There are  $n$  places and each passenger has its own place on the ticket. The problem is that the first passenger in the queue is very nervous, he rushes in the bus and takes one place at random. Each next passenger takes his seat according to the ticket if the seat is free. And if his own seat is occupied each passenger takes a random free seat.

Let  $p_n$  be the probability that the last passenger in the queue will get his seat.

- What is the initial conditions  $p_2$ ?
- Build the recurrence equation for  $p_n$ .
- Guess the solution!

**Exercise 5.3.** I have 100 safes. Each safe can be closed without a key, but requires a specific unique key to be opened. I randomly distribute 100 keys into 100 safes, one key per safe. I leave 5 safes open and I close other safes. So I have access to 5 keys and I can use them to open other safes and inside these safes there are more keys...

Let's denote by  $p_{kn}$  the probability that I can open all  $n$  safes, given that in the beginning there are  $k$  safes open and  $(n - k)$  safes closed.

- What are the initial conditions  $p_{0n}$  and  $p_{nn}$ ?
- Build the recurrence equation for  $p_{kn}$ .
- Guess the solution!

Let's denote by  $e_{kn}$  the expected number of open safes, given that in the beginning there are  $k$  safes open and  $(n - k)$  safes closed.

- What are the initial conditions  $e_{0n}$  and  $e_{nn}$ ?
- Build the recurrence equation for  $e_{kn}$ .
- Guess the solution!

**Exercise 5.4.** You play a the following lottery with the casino. There is a hat on the table. A fair dice is thrown until the game ends.

- If the dice shows one then you get +1 dollar immediately and the game continues.
- If the dice shows two then +2 dollars are added to the hat and the game continues.
- If the dice shows three then +3 dollars are added to the hat and the game continues.
- If the dice shows four then the sum in the hat goes back to casino and the game continues.
- If the dice shows five then the sum in the hat goes back to the casino and the game ends.
- If the dice shows six then the game ends and you receive the content of the hat.

Let  $e_n$  be your total expected payoff if there are  $n$  dollars in the hat now.

1. Build the recurrence equation for  $e_n$ .
2. Find the solution.
3. What is the price of one dollar in the hat in real dollars? What is the fair price to participate in this game?

## 6 Day 6. Generating functions

**Idea 3.** Any sequence of numbers

$$(a_0, a_1, a_2, a_3, \dots)$$

can be written as a generating function

$$g(t) = a_0 + a_1 t^1 + a_2 t^2 + a_3 t^3 + \dots$$

Other variants of a generating function may be also useful

$$m(t) = a_0 + a_1 \exp(t) + a_2 \exp(2t) + a_3 \exp(3t) + \dots$$

or

$$e(t) = a_0 + a_1 t^1/1! + a_2 t^2/2! + a_3 t^3/3! + \dots$$

or even

$$f(t) = a_0 + a_1 \cos(t) + a_2 \cos(2t) + a_3 \cos(3t) + \dots$$

**Definition 6.1.** We will denote by  $\text{coef}(g(t), t^k)$  the coefficient  $a_k$  before the term  $t^k$  in the expansion

$$g(t) = a_0 + a_1 t^1 + a_2 t^2 + a_3 t^3 + \dots + a_k t^k + \dots$$

**Exercise 6.1.** Master the extraction of coefficients!

- a) For the function  $g(t) = (1 + t)^{100}$  find  $\text{coef}(g, t^2)$ ,  $\text{coef}(g, t^{50})$ .
- b) For the function  $g(t) = 1/(1 - t)$  find  $\text{coef}(g, t^2)$ ,  $\text{coef}(g, t^{50})$ ,  $\text{coef}(g, t^{2026})$ .
- c) For the function  $g(t) = \frac{1}{t(t+1)}$  find  $\text{coef}(g, t^k)$ .
- d) For the function  $g(t) = \sqrt{1 - t^2}$  find  $\text{coef}(g, t^{100})$  and  $\text{coef}(g, t^{101})$ .
- e) For the function  $g(t) = (1 + x^5 + x^7)^{20}$  find  $\text{coef}(g, x^{17})$ .

**Exercise 6.2.** Do you understand how coefficients are related to derivatives?

- a) You know that  $g(0) = 5$ ,  $g'(0) = 7$ ,  $g''(0) = 12$ ,  $g'''(0) = 20$ . Find  $\text{coef}(g, 1)$ ,  $\text{coef}(g, 1)$ ,  $\text{coef}(g, 2)$ ,  $\text{coef}(g, 3)$ .
- b) Write the relation between  $\text{coef}(g, t^k)$  and  $k$ -th derivative,  $\frac{\partial^k}{\partial t^k} g(0)$ .
- c) For the function  $g(t) = 1/(1 - t)^p$  where  $p \in \mathbb{N}$  find  $\text{coef}(g, t^2)$ ,  $\text{coef}(g, t^{50})$ ,  $\text{coef}(g, t^k)$ .

**Idea 4.** How to discover the equation for the generating function?

- Represent the studied objects as vertices of a tree. Here  $a_n$  will be the number of objects with distance from the root equal to  $n$ .
- Encode objects as words, where  $a_n$  is the number of  $n$ -letter words. These words correspond to path from the root to the object.
- Start from recurrence equation for  $a_n$  and use sigma-boy approach.
- Start from recurrence equation for  $a_n$  and use two-way ticket approach.
- Apply first step approach to generating function in probability theory.

**Exercise 6.3.** Replacing the Taylor expansion by the closed formula!

a) Find the closed formula for

$$g(x) = 1 + x + x^2 + x^3 + x^4 + \dots$$

b) Find the closed formula for

$$g(x) = 1 + 7x + 7^2x^2 + 7^3x^3 + 7^4x^4 + \dots$$

c) Find a more compact formula for

$$g(x) = 1 + x + x^2 + x^3 + x^4 + \dots + x^{2026}.$$

d) Find a more compact formula for

$$g(x) = 1 + \binom{10}{1}x + \binom{10}{2}x^2 + \binom{10}{3}x^3 + \dots + \binom{10}{10}x^{10}.$$

e) Find the closed formula for

$$g(x) = 0 + x + 2x^2 + 3x^3 + 4x^4 + \dots$$

Hint: one may use differentiation or  $n$ -letter words approach.

f) Find the closed formula for

$$g(x) = 1 \cdot 2x^2 + 2 \cdot 3x^3 + 3 \cdot 4x^4 + \dots$$

g) Find the closed formula for

$$g(x) = \binom{2}{2}x^2 + \binom{3}{2}x^3 + \binom{4}{2}x^4 + \dots$$

h) Find the closed formula for

$$g(x) = 0 + x + 2^2x^2 + 3^2x^3 + 4^2x^4 + \dots$$

Hint: sum up some previous results

**Exercise 6.4.** Consider the rabbits story behind the Fibonacci numbers:

We start with one pair of young rabbits  $F_1 = 1$ . It takes one period of time for a pair of rabbits to mature. Every mature pair of rabbits in one period of time will give birth to a new young pair of rabbits.

For consistency with the recurrence equation assume that  $F_0 = 0$ .

- a) Draw the tree where each rabbit alive at time  $n$  is a node that is  $n$ -vertices apart from the root.
- b) Encode each rabbit alive at time  $n$  as  $n$ -letter word. Hint: two types of letters are enough.
- c) Find the equation for the generating function.
- d) Find the generating function.
- e) Using the generating function find

$$\sum_{k=0}^{\infty} F_k/100^k, \quad \sum_{k=0}^{\infty} kF_k/100^k.$$

- f) Represent 0.01010203050813213455 approximately as a simple fraction.
- g) Recover the formula for Fibonacci numbers using «sigma-boy» approach.
- h) Recover the formula for Fibonacci numbers using «two-way ticked» approach.

**Exercise 6.5.** Consider the recurrence equation

$$a_n = 3a_{n-1} - 1, \quad a_0 = 1.$$

- a) Find the equation for the generating function using «sigma-boy» method.
- b) Find the closed form of the generating function from the equation.
- c) Find the closed form of the generating function using «two-way ticket» method.
- d) Find the formula for  $a_n$  and the value  $a_{1000}$ .

**Exercise 6.6.** Consider the recurrence equation

$$a_n = 3a_{n-1} + 2n - 1, \quad a_1 = 4.$$

- a) Find  $a_0$  such that the recurrence formula works.
- b) Find the equation for the generating function using «sigma-boy» method.
- c) Find the closed form of the generating function from the equation.
- d) Find the closed form of the generating function using «two-way ticket» method.
- e) Find the formula for  $a_n$  and the value  $a_{1000}$ .

**Exercise 6.7.** Convolution!

Consider two generating functions

$$A(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots, \quad B(x) = b_0 + b_1x + b_2x^2 + b_3x^3 + \dots$$

and they product  $C(x) = A(x) \cdot B(x)$ .

- a) Find  $c_0, c_1, c_2, c_3$ .

- b) Find the general formula for  $c_k$ .

**Exercise 6.8.** Consider the recurrence equation

$$\binom{n+4}{4} = \sum_0^k a_k a_{n-k}, \quad a_0 = 1.$$

1. Find  $a_0, a_1, a_2$  by bare hands.
2. Find the equation for the generating function  $g(t)$ .
3. Find the formula for  $a_n$ .

### Home assignment: day 6 → day 9

1. Consider the recurrence equation

$$a_n = 3a_{n-1} + 5, \quad a_0 = 1.$$

- (a) Find equation for the generating function  $g(t)$ .
- (b) Obtain  $g(t)$  in a closed form.
- (c) Extract coefficients  $(a_n)$  from  $g(t)$ .

2. Extract the sequence  $(a_n)$  from the generating functions

$$g(t) = \frac{1}{(1-t)^{50}}, \quad g(t) = \frac{1}{(1-3t)^{100}}.$$

3. Find the generating function of the following sequences

$$(1, 67, 67^2, 67^3, 67^4, 67^5, \dots), \quad \left( \binom{5}{0}, \binom{6}{1}, \binom{7}{2}, \binom{8}{3}, \dots \right).$$

Hint: you don't need to solve equations, just recognize the patterns we have already seen.

4. You know that the generating function  $g(t)$  of some sequence  $(a_0, a_1, a_2, \dots)$  satisfies the equation

$$g(t) = 4 - 9t + (5t - 6t^2)g(t).$$

- (a) Find  $g(t)$ .
- (b) By simply plugging some value of  $t$  inside  $g(t)$  find the sum

$$\sum_{n=0}^{\infty} \frac{a_n}{100^n}.$$

### Demo for the quiz on day 7

1. Extract the sequence  $(a_n)$  from the generating functions:

$$g(t) = \frac{1}{1-3t}, \quad g(t) = \frac{1}{1+7t}, \quad g(t) = \frac{t}{1-3t}.$$

2. Decompose the fraction into a sum of more simple fractions

$$g(t) = \frac{4-9t}{(1-2t)(1-3t)}.$$

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## 7 Day 7. Moorrreee generating functions

**Exercise 7.1.** You can write any positive integer number on the sides of a dice.

- a) Is it possible to create two different non-standard dices  $Y_1$  and  $Y_2$  such that the distribution of  $Y_1 + Y_2$  is the same as the distribution of a sum of two fair dices  $X_1 + X_2$ ?
- b) Is it possible to create two identical non-standard dices  $Y_1$  and  $Y_2$  such that the distribution of  $Y_1 + Y_2$  is the same as the distribution of a sum of two fair dices  $X_1 + X_2$ ?
- c) Is it possible to create two different non-standard dices  $Y_1$  and  $Y_2$  such that the distribution of  $Y_1 + Y_2$  is uniform from 2 to 12?
- d) Is it possible to create two identical non-standard dices  $Y_1$  and  $Y_2$  such that the distribution of  $Y_1 + Y_2$  is uniform from 2 to 12?
- e) Is it possible to create one non-standard dice  $Y_1$  such that the distribution of sum with a standard second dice  $Y_1 + X_2$  is uniform from 2 to 12?

**Exercise 7.2.**

## 8 Day 8. Happy day ;)

## 9 Day 9. Generating functions in probability

**Exercise 9.1.** I throw a fair coin until the sequence  $HTH$  appears.

- a) Draw the directed graph for this game.
- b) Setup a system of equations for the sequences with the given endpoint and starting point at the start of the game.
- c) Setup a system of equations for the sequences with the given starting point and endpoint at the end of the game.
- d) Setup a system of equations for the non-terminal sequences with the given starting point.

Let  $N$  be the number of tosses,  $N_H$  — the number of  $H$  and  $N_T$  — the number of  $T$  till the end of the game.

- e) Find expected values  $\mathbb{E}(N)$ ,  $\mathbb{E}(N_H)$ ,  $\mathbb{E}(N_T)$ .
- f) Find probabilities  $\mathbb{P}(N = 12)$ ,  $\mathbb{P}(N_H = 13)$ ,  $\mathbb{P}(N_T = 14)$ .
- g) Find variances  $\mathbb{V}\text{ar}(N)$ ,  $\mathbb{V}\text{ar}(N_H)$ ,  $\mathbb{V}\text{ar}(N_T)$ .
- h) Find correlation  $\mathbb{C}\text{orr}(N_H, N_T)$ .

Note: you may use sympy from python or other software for symbolic computations.

**Exercise 9.2.** Alice and Bob throw a fair coin until the sequence  $A = HTH$  or the sequence  $B = HHT$  appears. Alice wins if the sequence  $A$  appears first, Bob wins if the sequence  $B$  appears first.

- a) Draw the directed graph for this game.
- b) Setup a system of equations for the sequences with the given endpoint and starting point at the start of the game.
- c) Setup a system of equations for the sequences with the given starting point and endpoint at the end of the game.
- d) Setup a system of equations for the non-terminal sequences with the given starting point.

Let  $N$  be the number of tosses,  $N_H$  — the number of  $H$  and  $N_T$  — the number of  $T$  till the end of the game.

- e) Find expected values  $\mathbb{E}(N)$ ,  $\mathbb{E}(N_H)$ ,  $\mathbb{E}(N_T)$ .
- f) What is the probability that Alice wins?
- g) Find probabilities  $\mathbb{P}(N = 12)$ ,  $\mathbb{P}(N_H = 13)$ ,  $\mathbb{P}(N_T = 14)$ .
- h) Find variances  $\text{Var}(N)$ ,  $\text{Var}(N_H)$ ,  $\text{Var}(N_T)$ .
- i) Find correlation  $\text{Corr}(N_H, N_T)$ .
- j) Find conditional probabilities  $\mathbb{P}(N = 12 \mid \text{Alice wins})$ ,  $\mathbb{P}(N_H = 13 \mid \text{Alice wins})$ ,  $\mathbb{P}(N_T = 14 \mid \text{Alice wins})$ .
- k) Find conditional expected values  $\mathbb{E}(N \mid \text{Alice wins})$ ,  $\mathbb{E}(N_H \mid \text{Alice wins})$ ,  $\mathbb{E}(N_T \mid \text{Alice wins})$ .

Note: you may use sympy from python or other software for symbolic computations.

**Exercise 9.3.**

**Exercise 9.4.**

**Exercise 9.5.**

## Home assignment: day 9 → day 12

You can use sympy or any other software for symbolic computations.

1. I throw a fair coin until the sequence  $HTHT$  appears.

- a) Draw the directed graph for this game.
- b) Setup a system of equations for the sequences with given endpoint and starting point at the start of the game.
- c) Setup a system of equations for the sequences with given starting point and endpoint at the end of the game.

Let  $N$  be the number of tosses,  $N_H$  — the number of  $H$  and  $N_T$  — the number of  $T$  till the end of the game.

- d) Find expected values  $\mathbb{E}(N)$ ,  $\mathbb{E}(N_H)$ ,  $\mathbb{E}(N_T)$ .
- e) Find probabilities  $\mathbb{P}(N = 12)$ ,  $\mathbb{P}(N_H = 13)$ ,  $\mathbb{P}(N_T = 14)$ .

2. Alice and Bob throw a fair coin until the sequence  $A = HHH$  or the sequence  $B = THH$  appears. Alice wins if the sequence  $A$  appears first, Bob wins if the sequence  $B$  appears first.

- a) Draw the directed graph for this game.
- b) Setup a system of equations for the sequences with the given endpoint and starting point at the start of the game.
- c) Setup a system of equations for the sequences with the given starting point and endpoint at the end of the game.

Let  $N$  be the number of tosses,  $N_H$  — the number of  $H$  and  $N_T$  — the number of  $T$  till the end of the game.

- d) Find expected values  $\mathbb{E}(N)$ ,  $\mathbb{E}(N_H)$ ,  $\mathbb{E}(N_T)$ .
- e) What is the probability that Alice wins?
- f) Find probabilities  $\mathbb{P}(N = 12)$ ,  $\mathbb{P}(N_H = 13)$ ,  $\mathbb{P}(N_T = 14)$ .

### Demo for the quiz on day 10

1. I throw a fair coin until the sequence  $HTHT$  appears. Draw the directed graph for this game.
2. Alice and Bob throw a fair coin until the sequence  $A = HHH$  or the sequence  $B = THH$  appears. Alice wins if the sequence  $A$  appears first, Bob wins if the sequence  $B$  appears first. Draw the directed graph for this game.

## 10 Day 10. Polya-Redfield counting

**Definition 10.1.** Necklace is a circle of  $n$  coloured beads that are considered equivalent under rotations.  
Cyclic group  $C_n$  is a collection of all possible rotations of a necklace.

**Definition 10.2.** Bracelet is a circle of  $n$  coloured beads that are considered equivalent under rotations and flips (turn-overs).

Dihedral group  $D_n$  is a collection of all possible rotations and flips of a bracelet.

We will work with a set  $A$  with  $\text{size}(A) = n$ . The number of colours will be denoted as  $k$ . We will denote the set of all colorings of  $A$  by  $X$ . The group  $G$  acts on  $A$  and therefore on  $X$  by permuting the elements.

**Definition 10.3.** A group of actions  $G$  should satisfy two properties:

1. For every action  $g$  there should exist an anti-action denoted by  $g^{-1}$ .
2. For every two actions  $a$  and  $b$  there should exist the combined action  $b \cdot a$ . Here  $a$  is applied first, and  $b$  is applied second. Convention is similar to  $\cos \sin(x)$  where  $\sin$ -function is applied first and then  $\cos$ -function is applied.

From these two principles it follows that every group  $G$  contains a «chill» action, identity element, denoted by  $e$ .

**Definition 10.4.** Consider an action  $g = (12)(3)(4)(56)(789)$ . The cyclic monomial of the action  $g$  is given by

$$\text{CM}_g(x_1, x_2, \dots, x_n) = x_2 \cdot x_1 \cdot x_1 \cdot x_2 \cdot x_3 = x_1^2 x_2^2 x_3.$$

In general the cyclic monomial of an action is the product

$$\text{CM}_g(x_1, x_2, \dots, x_n) = \prod_{i=1}^n x_i^{r_i}.$$

where  $r_i$  is the number of cycles of length  $i$  in the action  $g$ .

**Definition 10.5.** The cyclic index of the group  $G$  is the polynomial given by the scaled sum of cyclic monomials for every action

$$\text{CI}(x_1, x_2, \dots, x_n) = \frac{1}{\text{size}(G)} \sum_{g \in G} \text{CM}_g(x_1, x_2, \dots, x_n).$$

**Theorem 10.6** (Polya-Redfield theorem). The generating function for all colorings of the set  $A$  into three colors ( $a$ ,  $b$  and  $c$ ) different with respect to group  $G$  is given by

$$g(a, b, c) = \text{CI}(a + b + c, a^2 + b^2 + c^2, a^3 + b^3 + c^3, \dots, a^n + b^n + c^n).$$

**Theorem 10.7** (Burnside lemma). The number of colorings of the set  $A$  into  $k$  colors different with respect to group  $G$  is given by

$$\text{size}(X/G) = g(1, 1, \dots, 1) = \text{CI}(k, k, k, \dots, k) = \frac{1}{\text{size}(G)} \sum_{g \in G} \text{size}(\text{fix}(g, X)).$$

**Exercise 10.1.** Consider a necklace with 3 beads.

- How many different necklaces of  $k$  colours are there?
- List all the necklaces of 3 colors using a generating function.

**Exercise 10.2.** Consider a bracelet with 3 beads.

- How many different necklaces of  $k$  colours are there?
- List all the necklaces of 3 colors using a generating function.

**Exercise 10.3.** Consider a necklace with 4 beads.

- How many different necklaces of  $k$  colours are there?
- List all the necklaces of 3 colors using a generating function.

Note: you may use sympy or other software for symbolic computations.

**Exercise 10.4.** Consider a bracelet with 4 beads.

- How many different necklaces of  $k$  colours are there?
- List all the necklaces of 3 colors using a generating function.

Note: you may use sympy or other software for symbolic computations.

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**Exercise 10.5.** We consider two cubic dices identical if one is the rotation of the other.

- a) Explicitly describe the group of actions  $G$ .
- b) In how many ways can we label the faces of a dice with numbers from 1 to 6 if we should use every number exactly once?
- c) In how many ways can we label the faces of a dice with numbers from 1 to 3 if we should use every number exactly twice?
- d) Describe all possible dices with face labels from 1 to 3 using a generating function.

Note: you may use sympy or other software for symbolic computations.

**Exercise 10.6.** We consider two cubic dices identical if one is the rotation of the other.

- a) Explicitly describe the group of actions  $G$ .
- b) In how many ways can we paint the vertices of a cube in no more than  $k$  colors?
- c) In how many ways can we paint the vertices of a cube in exactly  $k$  colors?
- d) Describe all possible vertex colouring with no more than 3 colours using a generating function.

Note: you may use sympy or other software for symbolic computations.

**Exercise 10.7.** State the Polya-Redfield theorem for 4 colors.

## 11 Day 11. More Polya-Redfield counting

**Exercise 11.1.** We consider two tetrahedrons identical if one is the rotation of the other.

- a) Explicitly describe the group of actions  $G$ .
- b) In how many ways can we colour the faces of a tetrahedron with  $k$  colors?
- c) Describe all possible tetrahedrons with faces coloured in 3 colors using a generating function.

Note: you may use sympy or other software for symbolic computations.

**Exercise 11.2.** We consider two tetrahedrons identical if one is the rotation of the other.

- a) Explicitly describe the group of actions  $G$ .
- b) In how many ways can we gems of no more than  $k$  colors in the vertices of a tetrahedron?
- c) Describe all possible tetrahedrons with gems of 3 colors in the vertices using a generating function.

Note: you may use sympy or other software for symbolic computations.

**Exercise 11.3.** Let's consider a graph on 4 vertices. Two graphs are called isomorphic if they can be transformed one into another by reenumeration of vertices.

- a) Explicitly describe the group of actions  $G$ .
- b) In how many ways can we colour the edges of the complete graph  $K_4$  with no more than  $k$  colors?
- c) In how many ways can we colour the edges of the complete graph  $K_4$  using exactly  $k$  colors?
- d) Describe all possible graphs with 4 vertices and edges painted into no more than 3 colours using a generating function.
- e) How many non-isomorphic graphs with 4 vertices are there?

Note: you may use sympy or other software for symbolic computations.

**Exercise 11.4.** Let's consider a  $3 \times 3$  mosaic of no more than  $k$  colors. We consider two mosaics the same if one can be obtained from the other using rotations or flips.

- a) Explicitly describe the group of actions  $G$  acting on mosaics.
- b) How many mosaics are there?
- c) Describe all possible mosaics with no more than 3 colours using a generating function.

Note: you may use sympy or other software for symbolic computations.

**Exercise 11.5.** Count the number of final positions in  $3 \times 3$  noughts-and-crosses (tic-tac-toe) game.

### Home assignment: day 11 → day 14

1. Let's consider a  $3 \times 3$  mosaic of no more than  $k$  colors. We consider two mosaics the same if one can be obtained from the other using rotations or flips.
  - a) Explicitly describe the group of actions  $G$  acting on mosaics.
  - b) How many different mosaics are there?
  - c) Describe all possible mosaics with no more than 3 colours using a generating function.

Note: you may use sympy or other software for symbolic computations.

2. Watch the Mirek Olšák video «Burnside's lemma: counting up to symmetries» [Olš20].

[https://www.youtube.com/watch?v=D0d9bYZ\\_qDY](https://www.youtube.com/watch?v=D0d9bYZ_qDY),



3. Watch the Thom de Boer video «Burnside's lemma: An Introduction to Group Theory» [Boe23].

<https://www.youtube.com/watch?v=Exot1TqjEly>,



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## Demo for the quiz on day 12

Let's consider a  $4 \times 4$  mosaic of gems where each gem can be one of  $k$  types. Consider a  $90^\circ$  clockwise rotation of the mosaic.

1. Represent this rotation as a sequence of cycles and write the corresponding cycle monomial.
2. How many  $4 \times 4$  mosaics are fixed by this rotation?

## 12 Day 12. Planarity and Euler's formula

**Theorem 12.1** (Euler characteristic). If a polyhedra with  $v$  vertices,  $f$  faces and  $e$  edges is drawn on a surface with  $h$  holes then

$$v - e + f = 2 - 2h.$$

Note: the hole should be a part of the polyhedra!

**Definition 12.2.** Complete graph  $K_n$  consists of  $n$  vertices. All edges are present.

**Definition 12.3.** Complete multipartite graph  $K_{a,b,c,\dots}$  consists of groups of nodes. The first group has  $a$  vertices, the second one —  $b$  vertices, the third one —  $c$  vertices, and so on. There are no edges between vertices of the same group. All edges between members of different groups are present.

**Definition 12.4.** The graph  $B$  is called a subdivision of a graph  $A$  if graph  $B$  is obtained from graph  $A$  by adding vertices on already existing edges.

**Theorem 12.5** (Kuratowski's theorem). A graph is planar if and only if it contains no subdivision of  $K_{3,3}$  nor  $K_5$ .

**Exercise 12.1.** Prove the Euler characteristic formula. If a polyhedra with  $v$  vertices,  $f$  faces and  $e$  edges is drawn on a surface with  $h$  holes then

$$v - e + f = 2 - 2h.$$

**Exercise 12.2.** Consider a convex polyhedra with 20 vertices and 10 faces.  
How many edges does it have?

**Exercise 12.3.** Consider a plane connected graph with  $v$  vertices and  $e$  edges that divides the plane into  $f$  regions (faces).

- a) Find the value of the Euler characteristic  $v - e + f$ .
- b) How the answer will change if the graph has  $c$  components?

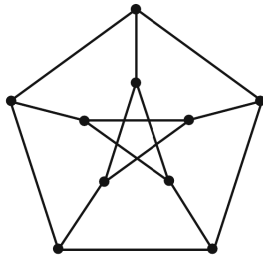
**Exercise 12.4.** Consider a football made of  $p$  pentagons and  $h$  hexagons with three faces meeting at each vertex.

- a) How many faces are there?
- b) How many vertices are there?
- c) How many edges are there?
- d) How many pentagons should be on a football?

**Exercise 12.5.** Draw  $K_5$ ,  $K_{3,3}$  and  $K_{2,2,2}$ .

Draw any subdivision of  $K_5$ ,  $K_{3,3}$  and  $K_{2,2,2}$ .

**Exercise 12.6.** Is the Peterson graph planar?



**Exercise 12.7.** a) Which complete graphs  $K_n$  are planar?

b) Which complete multipartite graphs  $K_{a,b}$  are planar?

c) Which complete multipartite graphs  $K_{a,b,c}$  are planar?

d) Which complete multipartite graphs  $K_{a,b,c,d}$  are planar?

e) Which complete multipartite graphs  $K_{a,b,c,d,e}$  are planar?

**Exercise 12.8.** Draw tetrahedron, cube, octahedron, dodecahedron, icosahedron.

**Exercise 12.9.** Consider a convex polyhedra.

a) What is bigger  $2e$  or  $3v$ ?

b) What is bigger  $2e$  or  $6f$ ?

c) What are possible values for the shortest face boundary (shortest cycle in the language of graph theory)?

**Exercise 12.10.** Draw tetrahedron, cube, octahedron, dodecahedron, icosahedron.

**Exercise 12.11.** Consider a regular polyhedra where each face is a  $k$ -gon and degree of every vertex is  $d$ .

a) What is bigger  $2e$  or  $dv$ ?

b) What is bigger  $2e$  or  $ke$ ?

c) Write the Euler characteristic identity in terms of  $d$ ,  $k$  and  $e$ .

d) What are the possible combinations of  $d$  and  $k$ ?

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### 13 Day 13. Graph colourings

### 14 Day 14. How to avoid combinatorics?

### 15 Day 15. Absolute happiness!

### 16 Yet to discover

#### Inclusions and exclusions

#### Proofs to DIE for

#### Wandermonde's identity

**Exercise 16.1.** Simplify

$$\binom{a}{0}\binom{b}{m} + \binom{a}{1}\binom{b}{m-1} + \binom{a}{2}\binom{b}{m-2} + \cdots + \binom{a}{m}\binom{b}{0}.$$
$$\binom{a}{0}\binom{m}{0} + \binom{a}{1}\binom{m}{1} + \binom{a}{2}\binom{m}{2} + \cdots + \binom{a}{m}\binom{m}{m}.$$
$$\binom{a}{0}^2 + \binom{a}{1}^2 + \binom{a}{2}^2 + \cdots + \binom{a}{a}^2.$$

**Exercise 16.2.** There are  $n = 500$  fishes in the pond,  $t = 30$  of them are tagged. You capture  $c = 5$  fishes at random without replacement. Let the random variable  $Y$  be the number of captured tagged fishes. Find the probabilities  $\mathbb{P}(Y = 0)$ ,  $\mathbb{P}(Y = 1)$ ,  $\mathbb{P}(Y = 2)$ ,  $\mathbb{P}(Y = y)$ .

**Exercise 16.3.** There are  $n = 24$  candies in the jar, 14 with orange flavour and 10 with ananas flavour. You take 6 candies at random. Let  $Y$  be the number of candies with orange flavour you picked. Find the probabilities  $\mathbb{P}(Y = 0)$ ,  $\mathbb{P}(Y = 1)$ ,  $\mathbb{P}(Y = 2)$ ,  $\mathbb{P}(Y = y)$ .

**Exercise 16.4.** A bridge hand (random 13 cards out of 52-card standard deck) is dealt.

- a) What is the probability that the hand contains no spades?
- b) What is the probability that the hand contains exactly 5 hearts?
- c) What is the probability that the hand contains at most two aces?

**Exercise 16.5.** Wandermonde-like identities in a funny notation. Here I intentionally omit the running index  $i$ , I write the first summation term and specify with  $\uparrow$  and  $\downarrow$  which terms goes up or down in the sum.

- a) Simplify and rewrite in a standard notation

$$\sum \binom{n}{0\uparrow} \binom{n\downarrow}{k}.$$

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b) Simplify and rewrite in a standard notation

$$\sum \binom{a\uparrow}{a} \binom{n\downarrow}{k}.$$

c) Simplify and rewrite in a standard notation

$$\sum \binom{b}{0\uparrow} \binom{n}{k\downarrow}.$$

**Exercise 16.6.** Simplify

$$\sum_{a+b+c=k} \binom{n_1}{a} \binom{n_2}{b} \binom{n_3}{c}.$$

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- [Olš20] Mirek Olšák. Burnside's lemma: counting up to symmetries. 2020. URL: [https://www.youtube.com/watch?v=D0d9bYZ\\_qDY](https://www.youtube.com/watch?v=D0d9bYZ_qDY). Counting 3 by 3 mosaics with Burnside lemma, 13 minutes.
- [Boe23] Thom de Boer. Burnside's lemma: An Introduction to Group Theory. 2023. URL: <https://www.youtube.com/watch?v=Exot1TqjEly>. Counting bracelets using Burnside lemma, author uses the term necklace instead of bracelet, 15 minutes.
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